

University of Frontier Technology, Bangladesh

Department of Internet of Things and Robotics Engineering (IRE)

Session: 2021-22

Course Title: Undergraduate Capstone

Course Code: IRE 400

Undergraduate Capstone Report

Capstone Report Overview

Capstone Title:	NirmalNode: Green IoT and Adaptive AI-Assisted Local Air Purification and Pollution Prediction System for Industrial Hotspots in Bangladesh
Capstone Semester:	January 2026, July 2026
Reporting Semester:	January 2026
Report Submission Date:	14 th September, 2026

Student Information

Student ID	Student Name
2201014	Aar Raisatunnesa Hridika
2201030	Md Rakib Hassan Dipu

Supervisor Information

Name	Md. Ashiqussalehin
Designation	Lecturer
Department	Internet of Things and Robotics Engineering
University	University of Frontier Technology, Bangladesh

University of Frontier Technology, Bangladesh

Meeting with Supervisor

Date	Day	Agenda	Supervisor's Signature
15-12-2025	Monday	Greeting with Supervisor	
22-12-2025	Monday	Capstone Topic / Title Discussion	
29-12-2025	Monday	Capstone Proposal Discussion	
05-01-2026	Monday	Present 2 Papers and Discussion on Those Papers	
12-01-2026	Monday	Present 3 Papers and Discussion on Those Papers	
19-01-2026	Monday	Present 5 Papers and Discussion on Those Papers	
26-01-2026	Monday	Present 4 Papers and Discussion on Those Papers	
02-02-2026	Monday	Present 6 Papers and Discussion on Those Papers	
09-02-2026	Monday	Present Those Papers Summary	
16-02-2026	Monday	Hardware Component Selection and Discussion	
23-02-2026	Monday	System Architecture Design Discussion	
02-03-2026	Monday	ESP32 Firmware and Sensor Integration Discussion	
09-03-2026	Monday	Presentation and Discussion on Tools We Learnt	
16-03-2026	Monday	AI/ML Model Selection and Methodology Discussion	

<u>Supervisor Name:</u> Md. Ashiqussalehin Lecturer Department of IRE, UFTB	<u>Signature and Date:</u>
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Meeting with Supervisor (Continued)

Date	Day	Agenda	Supervisor's Signature
23-03-2026	Monday	Prototype Hardware Assembly Review	
30-03-2026	Monday	Web Dashboard Design and Development Discussion	
06-04-2026	Monday	Data Collection and Field Testing Plan	
13-04-2026	Monday	AQI Calibration and Sensor Accuracy Discussion	
20-04-2026	Monday	Online ML Training Results Presentation	
27-04-2026	Monday	Methodology Presentation and Finalization	
04-05-2026	Monday	Capstone Writing — Chapter 1 & 2 Review	
11-05-2026	Monday	Capstone Writing — Chapter 3 & 4 Review	
18-05-2026	Monday	Capstone Writing — Chapter 5 & Results Discussion	
25-05-2026	Monday	Evaluation Metrics and Benchmark Comparison	
01-06-2026	Monday	Full Capstone Draft Review and Feedback	
08-06-2026	Monday	Preparation for Pre-Defense	
15-06-2026	Monday	Pre-Defense Presentation Practice	
29-06-2026	Monday	Post Pre-Defense Revision Discussion	

<u>Supervisor Name:</u> Md. Ashiqussalehin Lecturer Department of IRE, UFTB	<u>Signature and Date:</u>
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Meeting with Supervisor (Continued)

Date	Day	Agenda	Supervisor's Signature
06-07-2026	Monday	Draft Poster Design Discussion	
13-07-2026	Monday	Poster Finalization and Feedback	
20-07-2026	Monday	Final Capstone Revision and Proofreading	
27-07-2026	Monday	Defense Presentation Slide Review	
03-08-2026	Monday	Final Defense Practice Session	
10-08-2026	Monday	Capstone Formatting and Reference Check	
17-08-2026	Monday	Draft Poster and Other Documents Presentation	
24-08-2026	Monday	Final Revisions on All Chapters	
31-08-2026	Monday	Preparation of All Final Documents	
07-09-2026	Monday	Final Submission Document Verification	
14-09-2026	Sunday	Final Submission of All Documents	

<u>Supervisor Name:</u> Md. Ashiqussalehin Lecturer Department of IRE, UFTB	<u>Signature and Date:</u>
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